ETSI TS 146 053 V19.0.0 (2025-10)

TECHNICAL SPECIFICATION

Digital cellular telecommunications system (Phase 2+) (GSM); ANSI-C code for the GSM Enhanced Full Rate (EFR) speech codec (3GPP TS 46.053 version 19.0.0 Release 19)

Reference
RTS/TSGS-0446053vj00

Keywords
GSM

ETSI

650 Route des Lucioles
F-06921 Sophia Antipolis Cedex - FRANCE

Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16

Siret N° 348 623 562 00017 - APE 7112B
Association à but non lucratif enregistrée à la
Sous-Préfecture de Grasse (06) N° w061004871

Important notice

The present document can be downloaded from the
ETSI Search & Browse Standards application.

The present document may be made available in electronic versions and/or in print. The content of any electronic and/or print versions of the present document shall not be modified without the prior written authorization of ETSI. In case of any existing or perceived difference in contents between such versions and/or in print, the prevailing version of an ETSI deliverable is the one made publicly available in PDF format on ETSI deliver repository.

Users should be aware that the present document may be revised or have its status changed, this information is available in the Milestones listing.

If you find errors in the present document, please send your comments to the relevant service listed under Committee Support Staff.

If you find a security vulnerability in the present document, please report it through our Coordinated Vulnerability Disclosure (CVD) program.

Notice of disclaimer & limitation of liability

The information provided in the present deliverable is directed solely to professionals who have the appropriate degree of experience to understand and interpret its content in accordance with generally accepted engineering or other professional standard and applicable regulations.
No recommendation as to products and services or vendors is made or should be implied.
No representation or warranty is made that this deliverable is technically accurate or sufficient or conforms to any law and/or governmental rule and/or regulation and further, no representation or warranty is made of merchantability or fitness for any particular purpose or against infringement of intellectual property rights.
In no event shall ETSI be held liable for loss of profits or any other incidental or consequential damages.

Any software contained in this deliverable is provided "AS IS" with no warranties, express or implied, including but not limited to, the warranties of merchantability, fitness for a particular purpose and non-infringement of intellectual property rights and ETSI shall not be held liable in any event for any damages whatsoever (including, without limitation, damages for loss of profits, business interruption, loss of information, or any other pecuniary loss) arising out of or related to the use of or inability to use the software.

Copyright Notification

No part may be reproduced or utilized in any form or by any means, electronic or mechanical, including photocopying and microfilm except as authorized by written permission of ETSI.
The content of the PDF version shall not be modified without the written authorization of ETSI.
The copyright and the foregoing restriction extend to reproduction in all media.

© ETSI 2025.
All rights reserved.

Intellectual Property Rights

Essential patents

IPRs essential or potentially essential to normative deliverables may have been declared to ETSI. The declarations pertaining to these essential IPRs, if any, are publicly available for **ETSI members and non-members**, and can be found in ETSI SR 000 314: *"Intellectual Property Rights (IPRs); Essential, or potentially Essential, IPRs notified to ETSI in respect of ETSI standards"*, which is available from the ETSI Secretariat. Latest updates are available on the ETSI IPR online database.

Pursuant to the ETSI Directives including the ETSI IPR Policy, no investigation regarding the essentiality of IPRs, including IPR searches, has been carried out by ETSI. No guarantee can be given as to the existence of other IPRs not referenced in ETSI SR 000 314 (or the updates on the ETSI Web server) which are, or may be, or may become, essential to the present document.

Trademarks

The present document may include trademarks and/or tradenames which are asserted and/or registered by their owners. ETSI claims no ownership of these except for any which are indicated as being the property of ETSI, and conveys no right to use or reproduce any trademark and/or tradename. Mention of those trademarks in the present document does not constitute an endorsement by ETSI of products, services or organizations associated with those trademarks.

DECT™, **PLUGTESTS™**, **UMTS™** and the ETSI logo are trademarks of ETSI registered for the benefit of its Members. **3GPP™**, **LTE™** and **5G™** logo are trademarks of ETSI registered for the benefit of its Members and of the 3GPP Organizational Partners. **oneM2M™** logo is a trademark of ETSI registered for the benefit of its Members and of the oneM2M Partners. **GSM**® and the GSM logo are trademarks registered and owned by the GSM Association.

Legal Notice

This Technical Specification (TS) has been produced by ETSI 3rd Generation Partnership Project (3GPP).

The present document may refer to technical specifications or reports using their 3GPP identities. These shall be interpreted as being references to the corresponding ETSI deliverables.

The cross reference between 3GPP and ETSI identities can be found at 3GPP to ETSI numbering cross-referencing.

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in ETSI deliverables except when used in direct citation.

Contents

Intellectual Property Rights 2
Legal Notice 2
Modal verbs terminology 2
Foreword 4
1 Scope 5
2 References 5
3 Definitions and abbreviations 5
3.1 Definitions 5
3.2 Abbreviations 5
4 C code structure 6
4.1 Contents of the C source code disk 6
4.2 Program execution 6
4.3 Code hierarchy 7
Annex A (informative): Change History 13
History 14

Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

The present document provides the bit exact definition of the Enhanced Full Rate (EFR) speech traffic codec for the digital cellular telecommunications system.

An electronic attachment accompanies the present document, containing clause 5, the bit-exact ANSI-C code for the Enhanced Full Rate speech transcoder.

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

Version x.y.z

where:

- x the first digit:
  - 1 presented to TSG for information;
  - 2 presented to TSG for approval;
  - 3 or greater indicates TSG approved document under change control.
- y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.
- z the third digit is incremented when editorial only changes have been incorporated in the document.

1 Scope

The present document contains an electronic copy of the ANSI-C code for the GSM Enhanced Full Rate codec. The ANSI-C code is necessary for a bit exact implementation of the Enhanced Full Rate speech transcoder (GSM 06.60 [3]), Voice Activity Detection (GSM 06.82 [7]), comfort noise (GSM 06.62 [5]), Discontinuous Transmission (GSM 06.81 [6]) and example solutions for substituting and muting of lost frames (GSM 06.61 [4]).

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document.*

[1] GSM 01.04: "Digital cellular telecommunications system (Phase 2+); Abbreviations and acronyms".

[2] GSM 06.54: "Digital cellular telecommunications system (Phase 2+); Test sequences for the GSM Enhanced Full Rate (EFR) speech codec".

[3] GSM 06.60: "Digital cellular telecommunications system (Phase 2+); Enhanced Full Rate (EFR) speech transcoding".

[4] GSM 06.61: "Digital cellular telecommunications system (Phase 2+); Substitution and muting of lost frame for Enhanced Full Rate (EFR) speech traffic channels".

[5] GSM 06.62: "Digital cellular telecommunications system (Phase 2+); Comfort noise aspects for Enhanced Full Rate (EFR) speech traffic channels".

[6] GSM 06.81: "Digital cellular telecommunications system (Phase 2+); Discontinuous transmission (DTX) for Enhanced Full Rate (EFR) speech traffic channels".

[7] GSM 06.82: "Digital cellular telecommunications system (Phase 2+); Voice Activity Detector (VAD) for Enhanced Full Rate (EFR) speech traffic channels".

3 Definitions and abbreviations

3.1 Definitions

Definition of terms used in the present document can be found in GSM 06.60 [3], GSM 06.61 [4], GSM 06.62 [5], GSM 06.81 [6] and GSM 06.82 [7].

3.2 Abbreviations

For the purposes of the present document, the following abbreviations apply:

ANSI	American National Standards Institute
DS-HD	Double Sided High Density
ETS	European Telecommunication Standard
GSM	Global System for Mobile communications

I/O	Input/Output
ROM	Read Only Memory

For abbreviations not given in this clause see GSM 01.04 [1].

4 C code structure

This clause gives an overview of the structure of the bit-exact C code and provides an overview of the contents and organization of the archive en_300724v080000o0.ZIP which accompanies the present document.

The C code has been verified on the following systems:

- Sun Microsystems [1)] workstations and Sun Microsystems cc compiler and gcc compiler;
- IBM [2)] PC/AT compatible computers and Borland Turbo-C++ [3)] compiler;
- Hewlett Packard's [4)] workstations and HP cc compiler;

ANSI-C 9899 was selected as the programming language because portability was desirable.

4.1 Contents of the C source code disk

The C code disk has all of the files in the root level.

In this disk, the files with suffix "c" contain the source code and the files with suffix "h" are the header files. The ROM data is contained mostly in files with suffix "tab". All text files are formatted such that they are correct for an IBM PC/AT compatible.

The archive en_300724v080000o0.ZIP which accompanies the present document contains one speech coder installation verification data file, "spch_dos.inp". The reference encoder output file is named "spch_dos.cod", the reference decoder input file is named "spch_dos.dec" and the reference decoder output file is named "spch_dos.out". These four files are formatted such that they are correct for an IBM PC/AT compatible. The same files with reversed byte order of the 16 bit words are named "spch_unx.inp", "spch_unx.cod", "spch_unx."dec" and "spch_unx.out", respectively.

In an IBM PC/AT compatible platform, the installation verification can be performed by running the batch file "ts_dos.bat". In most UNIX platforms, the installation verification can be performed by running the batch file "ts_unx.bat". Final verification is to be performed using the GSM Enhanced Full Rate test sequences described in GSM 06.54 [2].

Makefiles are provided for the three platforms in which the C code has been verified (listed above). Once the software is installed, this directory will have compiled versions of *coder* and *decoder* (the bit-exact C executables of the speech codec), *ed_iface* (interface program between coder and decoder) and all the object files.

The programs *coder* and *decoder* are the GSM Enhanced Full Rate encoder and decoder executable files, respectively. A third program, *ed_iface*, is also contained in this directory. This is the program which provides the format conversion between the encoder output file format and the decoder input file format.

4.2 Program execution

The GSM enhanced full rate speech codec is implemented as three separate programs:

---

1) Registered trade mark of Sun Microsystems

2) Registered trade mark of International Business Machines

3) Registered trade mark of Borland

4) Registered trade mark of Hewlett Packard

- (*coder*) speech encoder;
- *(ed_iface)* encoder/decoder interface;
- (*decoder*) speech decoder.

For encoding using the *coder* program, the input is a binary speech file (*.inp) and the output is a binary encoded parameter file (*.cod). For decoding using the *decoder* program, the input is a binary parameter file (*.dec) and the output is a binary synthesized speech file (*.out).

NOTE: The format for the parameter input file required for decoding (*.dec) is not the same as the format of the parameter output file generated by encoding (*.cod). The *ed_iface* program will translate an *.cod file into a *.dec file.

See the file readme.txt for more information on how to run the coder, *ed_iface* and *decoder* programs.

4.3 Code hierarchy

Figures 1 to 5 are call graphs that show the functions used in the speech codec, including the functions of VAD, DTX, and comfort noise generation.

The encode call graph is broken down into three separate call graphs, and the decode call graph is broken down into two separate call graphs. Those clauses which are large are separated from the primary call tree and given their own call tree. Each vertical column represents a call level. For example, main() is at level 0, Coder_12k2() at level 1, Int_lpc2() at level 2, Lsp_Az() at level 3, Get_lsp_pol() at level 4, etc. The basic operations are not counted as extending the depth, therefore the deepest level in this software is level 4.

Some items have been omitted from this call graph. All standard C functions: printf(), fwrite(), etc. have been omitted. Also, no basic operations (add(), L_add(), mac(), etc.) or double precision extended operations (e.g. L_Extract()) appear in the graphs. The reset functions of the encoder and decoder are only visible as the functions reset_enc and reset_dec, respectively. There are several subroutine calls from inside these functions.

The time order in the call graphs is from the bottom upwards as the processing of a frame advances.

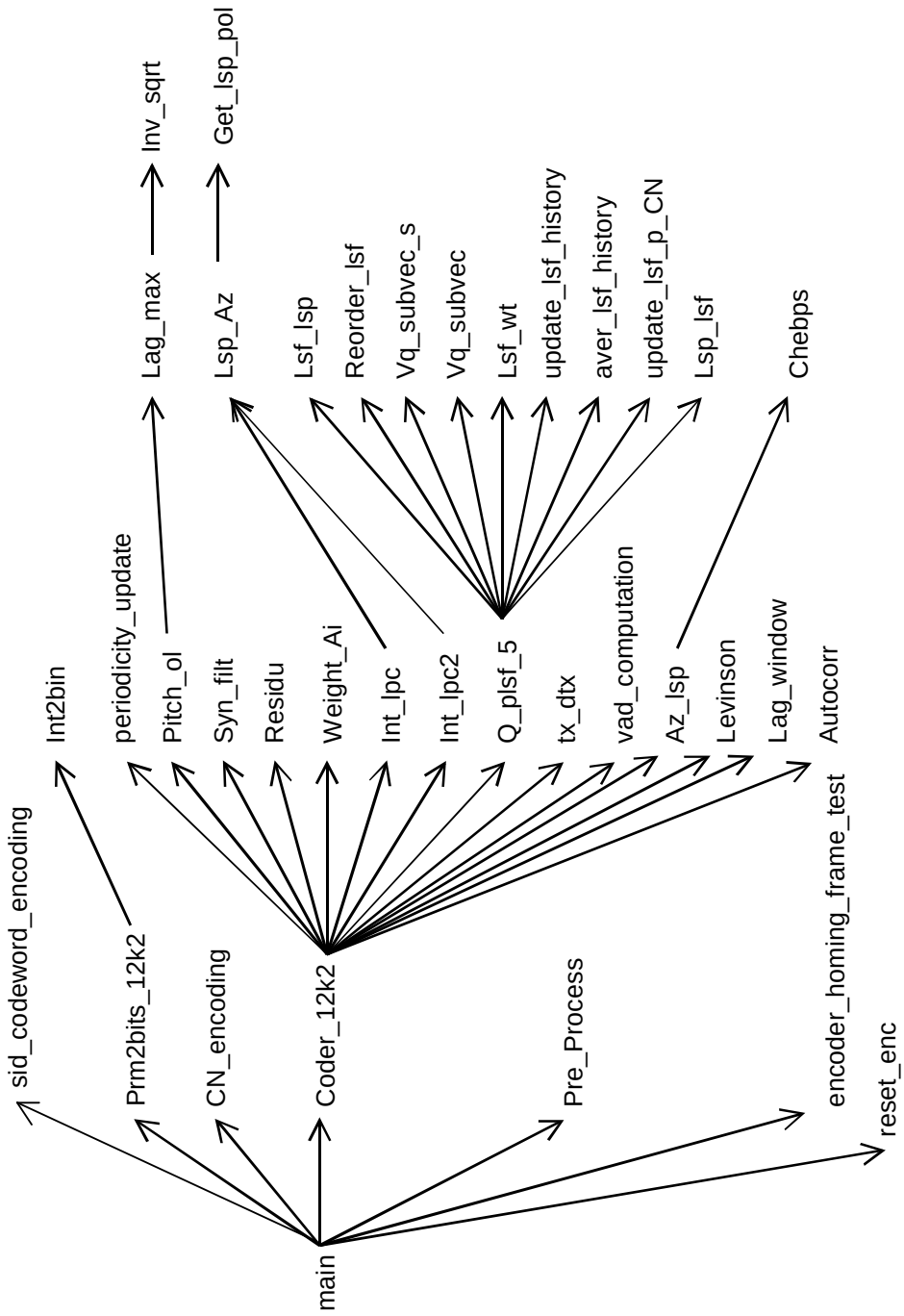

Figure 1: Speech encoder call graph (see figures 2 and 3)

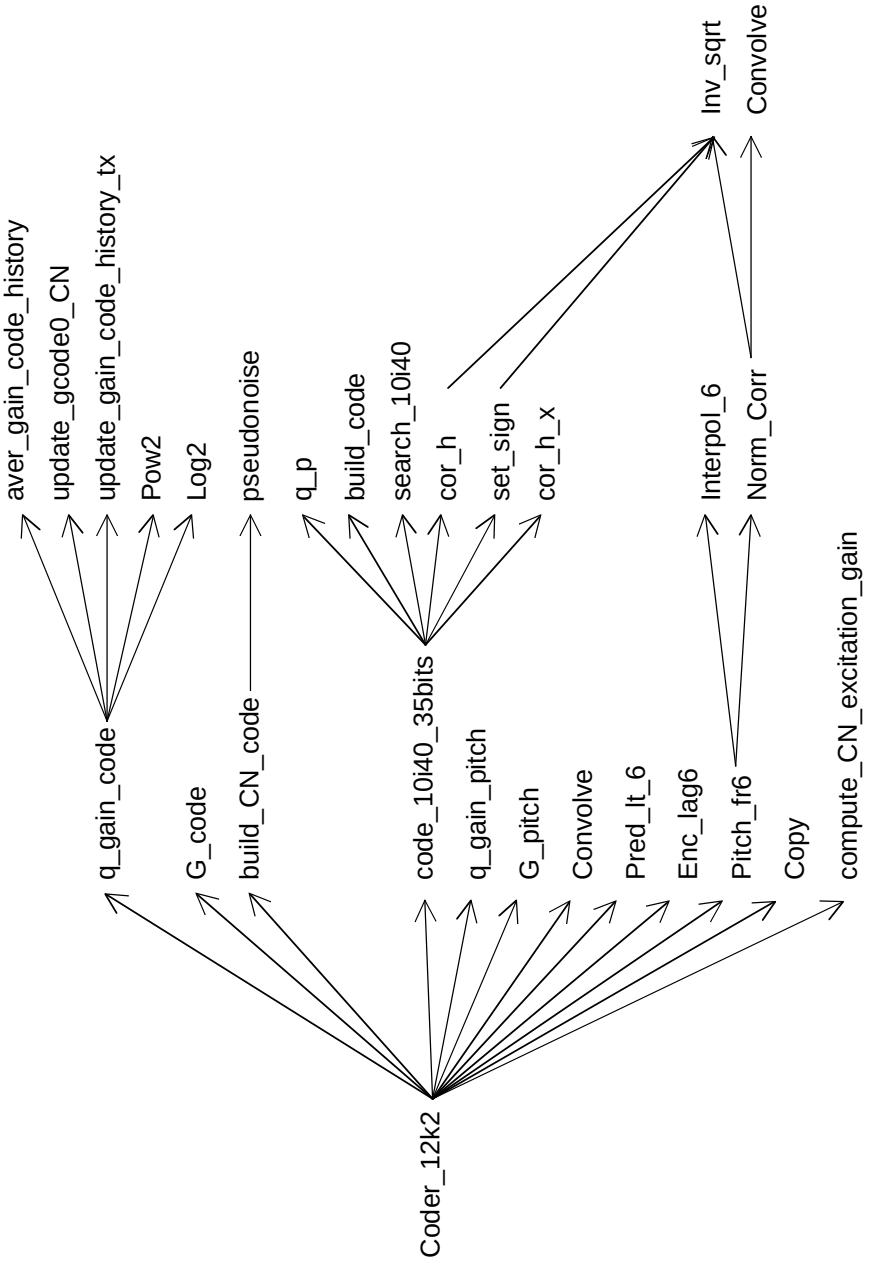

Figure 2: Speech encoder subframe processing call graph

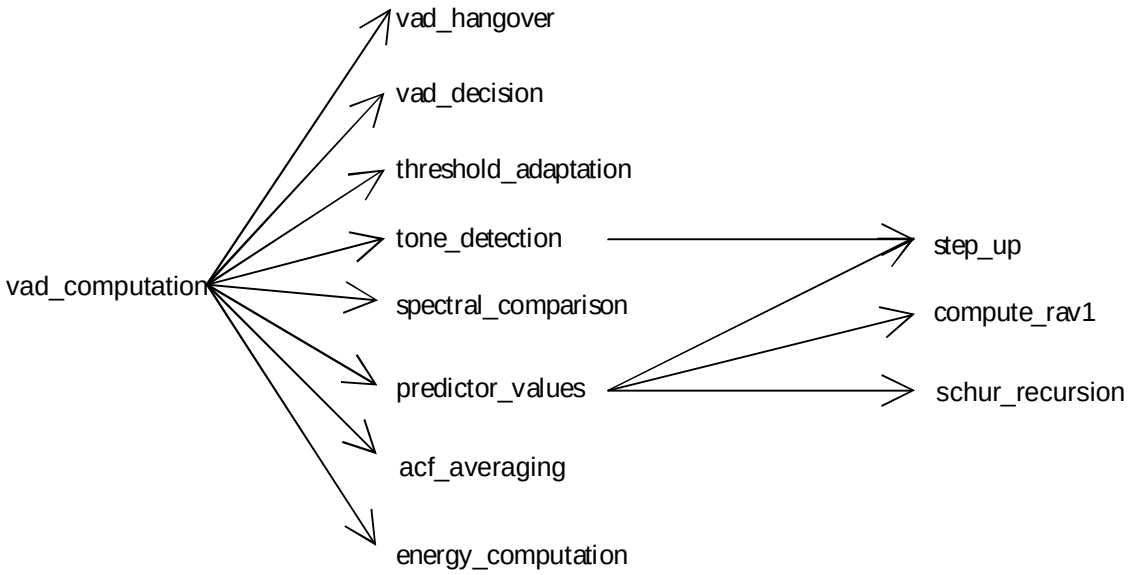

Figure 3: Voice Activity Detector (VAD) call graph

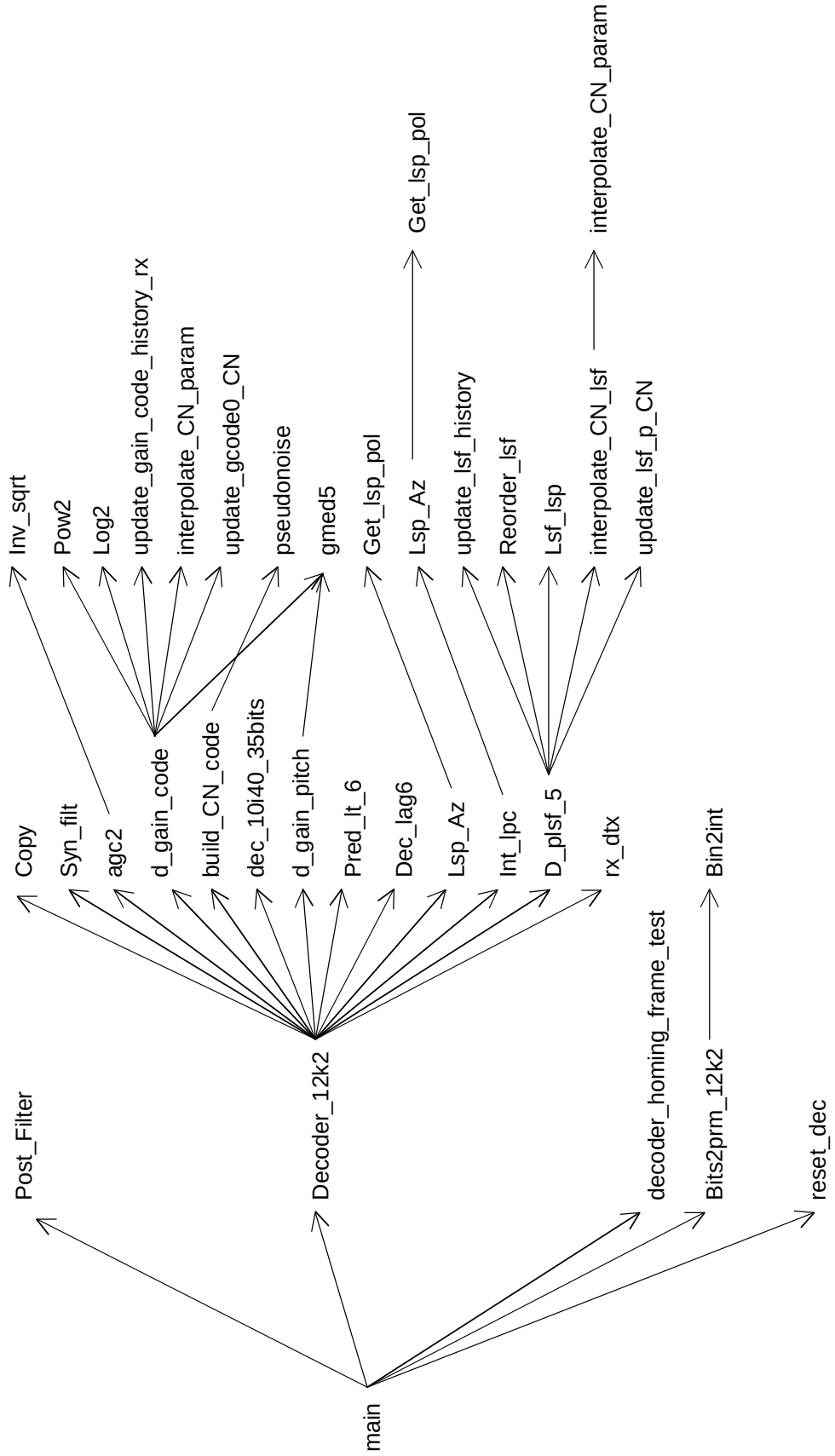

Figure 4: Speech decoder call graph (see figure 5)

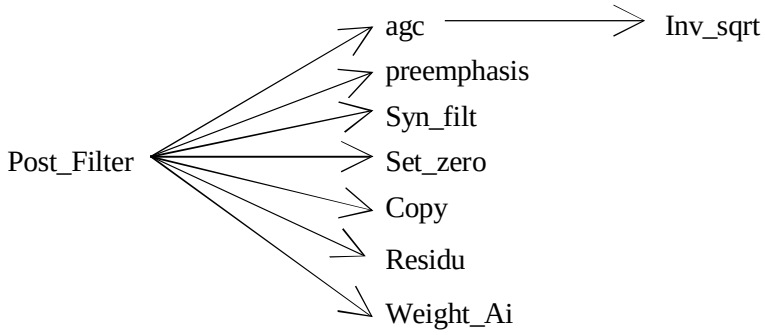

Figure 5: Speech decoder postfilter call graph

Annex A (informative): Change History

Change history					
SMG No.	**TDoc. No.**	**CR. No.**	**Clause affected**	**New version**	**Subject/Comments**
SMG#23				4.0.1	ETSI Publication
SMG#20				5.1.2	Release 1996 version
SMG#27				6.0.0	Release 1997 version
SMG#29				7.0.0	Release 1998 version
				7.0.1	Version update to 7.0.1 for Publication
SMG#31				8.0.0	Release 1999 version
				8.0.1	Update to Version 8.0.1 for Publication

Change history							
Date	**TSG #**	**TSG Doc.**	**CR**	**Rev**	**Subject/Comment**	**Old**	**New**
03-2001	11				Version for Release 4		4.0.0
06-2002	16				Version for Release 5	4.0.0	5.0.0
12-2004	26				Version for Release 6	5.0.0	6.0.0
06-2007	36				Version for Release 7	6.0.0	7.0.0
12-2008	42				Version for Release 8	7.0.0	8.0.0
12-2009	46				Version for Release 9	8.0.0	9.0.0
03-2011	51				Version for Release 10	9.0.0	10.0.0
09-2012	57				Version for Release 11	10.0.0	11.0.0
09-2014	65				Version for Release 12	11.0.0	12.0.0
12-2015	70				Version for Release 13	12.0.0	13.0.0

Change history							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
03-2017	SA#75					Version for Release 14	14.0.0
06-2018	SA#80	-	-	-	-	Version for Release 15	15.0.0
2020-07	-	-	-	-	-	Update to Rel-16 version (MCC)	**16.0.0**
2022-04	-	-	-	-	-	Update to Rel-17 version (MCC)	**17.0.0**
2024-03	-	-	-	-	-	Update to Rel-18 version (MCC)	**18.0.0**
2025-10	-	-	-	-	-	Update to Rel-19 version (MCC)	**19.0.0**

History

Document history		
V19.0.0	October 2025	Publication